

Decoding Visual Imagery from MEG Signals using Multi-Pathway Ensemble Learning

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Abstract

We address the task of decoding mentally imagined object categories from magnetoencephalography (MEG) brain recordings, submitted as a competitive entry to the IMAGINE Decoding Challenge on Kaggle (hosted by Dr. Gordon Feld, Central Institute of Mental Health, Mannheim). The competition requires classifying which of ten object categories - apple, bicycle, brush, cake, clown, cup, desk, foot, mountain, or zebra - a participant is imagining, given only stimulus-evoked (localizer) brain signals for training. The fundamental obstacle is a perception-to-imagery domain gap: MEG responses during active mental imagery are related to but neurologically distinct from those elicited by direct visual perception. We present a modality-aware multi-pathway ensemble that exploits a shared auditory processing window present in both paradigms. By training classifiers on the localizer auditory window (subjects hearing the word cue, approximately -1000 to 0 ms relative to visual onset) and applying them to the imagine auditory window (0 to 1000 ms, the same acoustic stimulus), we achieve same-modality transfer that substantially reduces the domain gap compared to cross-modal approaches. Our full system combines six weighted decoders spanning sliding-window one-vs.-rest logistic regression banks, full-window multinomial classifiers, and GloVe-300 word embedding regression. The ensemble achieves a Kaggle leaderboard accuracy of 0.133 (rank 27/169 teams), improving 23% relative over a baseline SVC at 0.108 (rank 59/92) in week 2.

1 Introduction

Recovering cognitive states from noninvasive neural recordings is a long-standing goal at the intersection of neuroscience and machine learning. Magnetoencephalography (MEG) measures the weak magnetic fields produced by synchronous neuronal currents at millisecond temporal resolution, making it uniquely suited for decoding the rapid dynamics of sensory and cognitive processing.

The IMAGINE Decoding Challenge.

This work describes our entry in the IMAGINE Decoding Challenge, a Kaggle competition hosted by Dr. Gordon Feld of the Central Institute of Mental Health, Mannheim, run as a course project for Johns Hopkins University's graduate-level Deep Neural Networks course (DNN 605.742, Spring 2026). The competition presents 169 teams with a 10-class object classification problem. Prizes of \$600/\$300/\$100 are awarded only if accuracy exceeds 20%; chance is 10%. No team surpassed the prize threshold.

Task definition.

Participants viewed or heard 10 object words (apple, bicycle, brush, cake, clown, cup, desk, foot, mountain, zebra) during

two phases. In the localizer phase, they heard an auditory cue then saw the corresponding object image. In the imagine phase, they heard only the auditory cue and mentally visualized the object with eyes closed. Localizer recordings carry ground-truth labels; imagine labels are scrambled and must be recovered by the model.

Problem statement.

Let X_{loc} in $\mathbb{R}^{\text{C} \times \text{T}_{\text{loc}}}$ denote a localizer epoch with label y in $\{1, \dots, 10\}$, and X_{img} in $\mathbb{R}^{\text{C} \times \text{T}_{\text{img}}}$ an imagine epoch. The goal is to learn f from the labeled localizer distribution $p(X_{\text{loc}}, y)$ that generalizes to $p(X_{\text{img}})$. The key challenge is:

$$p(X_{\text{loc}}) \neq p(X_{\text{img}}) \quad (1)$$

Neural dynamics during imagery only partially overlap with those during perception (Dijkstra et al., 2019; Bezsudnova et al., 2024).

Contributions.

We make three primary contributions: (1) identifying and exploiting a same-modality auditory transfer pathway that eliminates the perception-imagery domain gap for the acoustic segment; (2) constructing a six-decoder weighted ensemble combining auditory and visual pathways, sliding-window and full-window classifiers, and semantic embedding regression; (3) systematic analysis of two failed approaches distinguishing transfer failure from featurization failure.

2 Dataset and Experimental Setup

2.1 MEG Recording System

All recordings were collected using a MEGIN Triux system (formerly Elekta Neuromag) with a 306-channel Vectorview layout: 102 magnetometers and 204 planar gradiometers, plus ECG and EOG reference channels. Raw signals were downsampled to 100 Hz ($\Delta t = 0.01$ s) before release. Data are stored in MNE-Python Epochs format (one epoch = one trial). Signal Space Separation (SSS; Taulu & Simola, 2006) was applied at acquisition to suppress external magnetic interference.

2.2 Dataset Structure

The dataset comprises 29 subjects: 15 training, 14 test. Each contributes localizer epochs (480 trials, 48/class, -200 to +1000 ms re. visual onset, 121 steps) and imagine epochs (50 trials, -200 to +5000 ms re. auditory cue onset, 521 steps). Only localizer data may be used for training; imagine test labels are withheld.

Split	Type	Epochs/subj.	Channels	Time steps	Labels
Train (15)	Localizer	480 (48/class)	309	121	Provided

Train (15)	Imagine	50	309	521	Provided (local CV)
Test (14)	Localizer	480 (48/class)	309	121	Provided
Test (14)	Imagine	50	309	521	Scrambled - predict

Table 1. Dataset dimensions per subject. Localizer: -200 ms to +1000 ms re. visual onset. Imagine: -200 ms to +5000 ms re. auditory cue onset.

2.3 Computational Resources

All experiments ran on free-tier Google Colab: Intel Xeon @ 2.2 GHz (2 cores), 12.7 GB RAM, 78 GB storage. No GPU was used; all models are CPU-based scikit-learn classifiers. The absence of GPU acceleration was not limiting given our feature-based decoder architecture.

3 Preprocessing

3.1 Sensor Selection: Gradiometers Only

The MEGIN Triux layout provides 102 magnetometers and 204 planar gradiometers. Magnetometers are sensitive to deep and distant neural sources but also pick up far-field interference, increasing inter-subject variability. Planar gradiometers measure the spatial gradient of the magnetic field and are sensitive to local, superficial cortical sources - precisely the population-level activity we expect to discriminate between word categories. An ablation study comparing all 306 channels, gradiometers only, and magnetometers only confirmed that gradiometers alone produced the highest cross-validated accuracy.

3.2 Time Window Selection

Localizer auditory window.

The localizer epoch begins with the subject hearing the target word. Auditory cortex responds within 100 ms and sustains activation through ~400 ms (Hickok & Poeppel, 2007). This response is embedded in the localizer's negative-time segment (~-1000 to -50 ms re. visual onset, since the auditory cue precedes image presentation by ~1000 ms).

Localizer visual window.

Visual object processing produces category-discriminative responses from ~200 ms post-onset, peaking at 300-600 ms. We use +50 ms to +600 ms, excluding the early nonspecific onset response (0-200 ms) that is identical across all 10 categories.

Imagine windows.

The imagine epoch is locked to auditory cue onset ($t=0$). The auditory response occurs in 0-1000 ms, exactly as in the pre-visual localizer segment. Mental imagery emerges later, typically after 500 ms. We use 0-1000 ms for auditory-pathway decoders and 300-3000 ms for

3.3 ICA Artifact Removal: Tried and Rejected

ICA was evaluated to remove ECG and EOG artifacts (Jung et al., 2000). Models trained on ICA-cleaned data consistently performed worse due to: (1) signal loss - components correlated with ECG/EOG can carry genuine cortical activity, especially in temporal and frontal channels (Castellanos & Makarov, 2006); (2) redundancy - the MEGIN SSS preprocessing already attenuates external interference (Taulu & Simola, 2006). ICA was excluded from the final pipeline.

3.4 Null Data Injection

Pre-stimulus baseline epochs (-200 ms to 0 ms) are randomly sampled and assigned a null class label during OvR classifier training. This sharpens the decision boundary between genuine neural responses and resting-state activity, reducing false positives when the sliding-window decoder scans the imagery epoch at non-signal time points (Blankertz et al., 2011). Null injection increased validation accuracy by 1-2 points.

4 Feature Engineering

Three feature families are computed from the 204-channel gradiometer data, each capturing a distinct aspect of the spatiotemporal MEG signal structure.

4.1 Temporal Window Statistics

The analysis epoch is divided into $N_{win} = 5$ equal temporal segments. For each segment k and channel c , the mean and standard deviation of raw signal amplitude are computed:

$$\mu_{(c,k)} = (1/|W_k|) * \sum_{t \in W_k} x_{(c,t)} \quad (2)$$

$$\sigma_{(c,k)} = \sqrt{(1/|W_k|) * \sum (x_{(c,t)} - \mu)^2} \quad (3)$$

This yields $2 \times 5 \times 204 = 2,040$ features per epoch. The five windows provide coarse tracking of evoked response temporal dynamics while the standard deviation captures within-window variability.

4.2 Band-Power Features

For each of four frequency bands, we apply a 4th-order Butterworth bandpass filter, compute the Hilbert analytic envelope, and average power over the analysis window per channel:

$$BP_{(c,b)} = (1/T) * \sum_t |H\{x_{(c,t)}^{(b)}\}|^2 \quad (4)$$

This produces $4 \times 204 = 816$ additional features.

Band	Range (Hz)	Cognitive association
Theta	4 - 8	Working memory, object recognition

Beta	13 - 30	Motor planning, cognitive effort
Gamma	30 - 45	Local computation, feature binding

Table 2. Frequency bands for band-power extraction.

4.3 GloVe Word Embeddings

Each of the 10 target word categories is represented by its 300-dimensional GloVe vector (Pennington et al., 2014), pre-trained on Wikipedia + Gigaword (glove-wiki-gigaword-300). These embeddings encode inter-word semantic relationships: semantically related words have higher cosine similarity (e.g., 'apple' and 'cake' are both foods; 'mountain' and 'zebra' are unrelated).

4.4 Riemannian Covariance (Attempted, Excluded)

Riemannian tangent-space features (Ledoit-Wolf regularized 204x204 covariance matrices projected to the SPD manifold via pyriemann; Barachant et al., 2012) achieved mean 3-fold CV accuracy 0.238 on the localizer alone, substantially above chance. However, when applied to imagery, accuracy collapsed to 0.095 (range 0.020-0.160), indistinguishable from chance. The 14-point gap confirms that spatial covariance structure during mental imagery does not match that during visual perception. Excluded from the final ensemble (see Section 9).

5 Model Development Journey

Week 2 - Full MEG baseline.

Beginning from a minimal EOG-only baseline, we expanded to 306-channel MEG with Ridge-based artifact removal, 50 ms windowed averages (~3,264 features), RobustScaler, and SVC with RBF kernel (C=10). Validation: 10.93% (p=0.17 vs. chance). LB: 0.108, rank 59/92.

Week 3 - Channel ablation and window tuning.

Restricted to 204 gradiometers and revised the time window to exclude the first 200 ms (category-nonspecific auditory onset). Validation: 12.13% (p=0.077). LB: 0.114, rank 46/103.

FPIP - Feature engineering.

Implemented temporal window statistics, band-power features, null data injection, and per-time-window classifiers. LB: 0.120, rank 36/118.

Week 10 - Auditory pathway discovery.

The breakthrough insight - the same-modality transfer pathway through the shared auditory window (Section 6) - was discovered during week 10. Implementing the 6-decoder ensemble immediately improved the LB from 0.120 to 0.133, rank 16/142. This single architectural change was responsible for the majority of total improvement.

The two-phase localizer epoch.

The localizer epoch spans -200 ms to +1000 ms relative to visual image onset. This means the epoch contains not only the visual evoked response (0 ms onward) but also a substantial pre-visual period during which the subject hears the target word. The auditory word presentation occurs ~1000 ms before the image, so the auditory response is embedded in the negative-time portion of the localizer epoch.

The imagine epoch begins with the same stimulus.

In the imagine paradigm, subjects hear the same spoken word cue (0 ms onset), then visualize the object. Thus the imagine epoch's initial 0-1000 ms captures the brain's response to an identical acoustic stimulus as the localizer epoch's pre-visual period.

Why this eliminates the domain gap.

Prior approaches train on the localizer's visual window and apply to the imagery period - cross-modal transfer between two distinct brain processes. By contrast, training on the localizer's auditory window and testing on the imagine auditory window is same-modality transfer: both involve the auditory cortex processing the same acoustic input. The neural response to hearing 'apple' during localizer and imagine trials should be neurologically equivalent, modulo minor context effects. Formally, let ϕ_{aud} denote the auditory neural response. We hypothesize:

$$p(\phi_{\text{aud}}^{\text{loc}} | y) \approx p(\phi_{\text{aud}}^{\text{img}} | y) \quad (5)$$

This is not true for visual/imagery responses, where $p(\phi_{\text{vis}}^{\text{loc}} | y) \neq p(\phi_{\text{imagery}}^{\text{img}} | y)$. The hypothesis is supported empirically: auditory sliding OvR achieves ~14.1% mean validation accuracy vs. ~11.2% for visual sliding OvR, consistent with the reproducibility of auditory cortex responses to spoken words across contexts (Hickok & Poeppel, 2007).

7 Model Architecture: Six-Decoder Ensemble

The final system combines six independent decoders, each outputting a probability vector $p^{(d)}$ in $\mathbb{R}^{10}$ over the ten categories. The final prediction is:

$$\hat{y} = \underset{c}{\operatorname{argmax}} \sum_{d=1}^6 w_d \cdot p_c^{(d)} \quad (6)$$

where w_d is the weight for decoder d , assigned based on expected transfer quality.

Decoder	Train window	Apply window	w_d	Transfer type
Aud. Sliding OvR	Loc. aud. (-1s to 0s)	Img. (0-1s)	3.0	Same-modal
Aud. Full-Window	Loc. aud. (-1s to 0s)	Img. (0-1s)	2.0	Same-modal
Aud. Emb. Reg.	Loc. aud. (-1s to 0s)	Img. (0-1s)	1.5	Same-modal

Vis. Sliding OvR	Loc. vis. (0-0.6s)	Img. (0.3-3s)	1.0	Cross-modal
Vis. Full-Window	Loc. vis. (0-0.6s)	Img. (0.3-3s)	0.5	Cross-modal
Vis. Emb. Reg.	Loc. vis. (0-0.6s)	Img. (0.3-3s)	0.5	Cross-modal

Table 3. Six-decoder ensemble configuration. Auditory decoders receive higher weights due to same-modality transfer quality.

7.1 Sliding Window One-vs.-Rest Bank

A separate One-vs.-Rest (OvR) L1-penalized logistic regression is trained at every 10 ms time step across the training window (50 ms window width). For each time step t , the 50 ms x 204-channel segment is flattened to a 10,200-dimensional vector. For each class c in $\{1, \dots, 10\}$, a binary logistic regression is trained:

$$\min_{\{w, b\}} ||w||_1 + C \sum_i \log[1 + \exp(-y_i (w^T x_i + b))]$$

N_{null} null-class samples drawn from the pre-stimulus baseline are added with label -1 for all classes (null injection, Section 3.4). L1 penalty ($C=6.0$) encourages sparse channel/time weights.

Cross-validation pruning.

Each time window is evaluated by 3-fold CV. Windows where CV accuracy falls at or below 11% are discarded. This retains only time points with genuine class-discriminating signal, reducing inference noise.

Null logit z-scoring.

At inference, trial i 's logit at window t is z-scored against the null distribution estimated from baseline training samples, then centered per trial:

$$z(i, t, c) = (\text{logit}(i, t, c) - \mu_c^{\text{null}}) / \sigma_c^{\text{null}}$$

Aggregation.

The final probability estimate for decoder d is:

$$p^{(d)}_i = \text{softmax}(\sum_{t \in T_{\text{keep}}} \text{cv_acc}_t * z(i, t, \theta))$$

7.2 Full-Window Classifier

Temporal window statistics and band-power features (up to 2,856 dimensions) are reduced to 50 PCA components, then multinomial logistic regression (LBFGS, $C=1.0$, balanced class weights) is trained. At inference, the imagery epoch is scanned with 0.2 s steps; predicted probabilities are averaged over all scan positions.

7.3 Word Embedding Regression

A Ridge regression ($\alpha=100$) is trained to predict the 300-dimensional GloVe vector of the target class from PCA-reduced MEG features ϕ_i in $\mathbb{R}^{50}$:

$$\min_W ||W\phi_i - e_{yi}||^2 + \alpha ||W||_F^2 \quad (10)$$

For a test trial, the predicted embedding $\hat{e}$ is compared to all 10 target word embeddings by cosine distance:

A temperature-scaled softmax (tau=0.2) converts distances to probabilities:

p_c = exp(-d_c/tau) / sum_c' exp(-d_c'/tau), tau=0.2

8 Results

8.1 Competition Leaderboard Trajectory

The competition began with a baseline SVC at 0.108 and ended at 0.133 following the ensemble implementation in week 10.

Module	Score	Rank / Teams	Cum. Sub.	Key change
Week 2	0.108	59 / 92	2	Full MEG + SVC baseline
Week 3	0.108	65 / 100	4	-
Week 4	0.114	46 / 103	13	Gradiometers only
Week 5	0.114	46 / 107	21	Time window tuning
FPIP	0.120	36 / 118	24	Null injection + features
Week 9	0.120	41 / 132	26	-
Week 10	0.133	16 / 142	27	Auditory ensemble
Week 11	0.133	20 / 151	30	Ensemble tuning
Week 13	0.133	27 / 169	34	Final

Table 4. Weekly Kaggle leaderboard performance. Best rank 16th (week 10); final rank 27th/169.

8.2 Ablation Study: Individual Decoder Performance

Each decoder was evaluated in isolation on 15 training subjects using within-subject cross-validation (localizer -> imagine label accuracy).

Decoder	Mean acc.	Median
Aud. Sliding OvR	0.141	0.140
Aud. Full-Window	0.128	0.120
Aud. Emb. Reg.	0.119	0.100
Vis. Sliding OvR	0.112	0.100
Vis. Full-Window	0.108	0.100
Vis. Emb. Reg.	0.106	0.100
6-Decoder Ensemble	0.162	0.160

Table 5. Per-decoder and ensemble CV accuracy. Ensemble (0.162) substantially outperforms every individual component.

The ensemble substantially outperforms every individual decoder, confirming genuine complementarity between components. The auditory sliding OvR is the strongest individual decoder (0.141), consistent with the same-modality transfer hypothesis.

9 Failure Mode Analysis

Two decoder architectures showed strong within-domain

9.1 Riemannian Tangent-Space Decoder: Transfer Failure

Per-subject 3-fold CV accuracy on the localizer alone was 0.238 (range 0.146-0.450), well above chance. This confirms the decoder learns genuinely class-discriminating information: spatial covariance patterns during visual perception differ reliably across the 10 word categories.

When this localizer-trained decoder was applied to imagery, accuracy collapsed to 0.095 (range 0.020-0.160), indistinguishable from chance. The 14-point gap cannot be attributed to dimensionality (feature-to-sample ratio 20,910:480). The decoder can learn; it just cannot transfer across brain states. We attempted three remediation strategies:

1. **Covariance recentering:** subtracting the task-specific mean covariance before tangent projection. Did not recover usable accuracy.
2. **Occipital channel subset:** restricting covariance computation to occipito-temporal channels most involved in visual processing. Did not improve imagery accuracy beyond chance.
3. **Imagine anchoring:** mixing held-out imagery trials into the tangent space reference. Conceptually sound but requires using imagine labels for training, which the competition prohibits.

All three strategies failed. Riemannian features were excluded from the final ensemble.

9.2 Alpha-Band Log-Variance Decoder: Featurization Failure

We hypothesized that mental imagery involves sustained alpha-band desynchronization (8-13 Hz). An alpha-band decoder using 4th-order Butterworth filtering + channel-wise log-variance of the Hilbert envelope (1.0-3.0 s post-cue) achieved cross-task accuracy 0.084. To determine whether transfer was the limiting factor, we ran 5-fold CV entirely within imagery training data: accuracy 0.088 - still at chance. The two numbers being nearly identical ($0.084 \approx 0.088$) rules out the transfer hypothesis: the featurization simply contains no class-discriminating information. Three potential explanations: (1) wrong band - suppression may manifest in beta or lower gamma; (2) collapsed time dimension - log-variance over 1.0-3.0 s discards temporal dynamics that a time-resolved approach would preserve; (3)

effect too weak or spatially diffuse to discriminate between 10 categories at single-trial level. The alpha decoder was excluded from the final ensemble.

9.3 Subject-Level Prediction Collapse

Subject sub-26 exhibited degenerate predictions: 48 of 50 imagine trials were classified as 'zebra'. This failure mode occurs in per-subject OvR classifiers when one class's training distribution overlaps strongly with the test distribution for a particular subject's neural geometry. Future mitigation: cross-subject regularization constraining per-subject classifiers toward the population mean.

10 Discussion

10.1 The Primacy of Domain Shift

The most important lesson from this competition is that domain shift dominates over model complexity. The improvement from a simple SVC baseline (0.108) to the full ensemble (0.133) did not come from a more powerful classifier. It came from identifying a structural property of the experimental paradigm - the shared auditory window - that allowed same-modality transfer. The Riemannian experiment makes this point sharply: a decoder achieving 0.238 CV accuracy on the training domain collapses to 0.095 when transferred, while a simple logistic regression on temporal features achieves 0.141 on the target domain. Feature complexity is secondary to distribution alignment.

10.2 Semantic Embeddings as Complementary Signal

The GloVe embedding regression contributes a qualitatively different type of information: semantic structure. For confusable pairs such as 'cup' and 'desk' (both indoor objects) or 'apple' and 'cake' (both foods), the embedding regression distributes probability mass toward semantically nearby categories rather than random ones. This is consistent with the semantic brain decoding literature (Mitchell et al., 2008).

10.3 Ensemble Diversity

The 56% relative improvement of the ensemble (0.162) over the best individual decoder (0.141) demonstrates genuine complementarity between components. The three decoder families - sliding-window OvR (time-resolved discriminative), full-window (global temporal integration), and embedding regression (semantic) - each capture distinct aspects of the MEG signal.

10.4 Limitations

- Accuracy ceiling: Even the best ensemble (0.133) is well below the 0.20 prize threshold and far short of

require imagery-domain supervision to bridge reliably.

- Per-subject fitting: All classifiers are trained separately per subject, requiring ≥ 480 labeled localizer trials per subject. Cross-subject generalization is unaddressed.
- CPU-only: The sliding-window bank requires fitting hundreds of logistic regression models per subject per pathway - several minutes per subject. GPU-accelerated deep learning (EEGNet, BrainBERT) could provide richer representations.
- No temporal deep learning: Feature engineering collapses temporal dynamics to summary statistics. Recurrent or attention-based models could capture higher-order dynamics directly from the raw time series.

11 Future Work

Fine-tuning on imagine-train labels.

Using a small subset of labeled imagine trials to adapt localizer-trained classifiers via domain adaptation or few-shot fine-tuning. Even 5-10 labeled imagine trials per class could substantially reduce the domain gap.

Cross-subject models.

Training a single model across all 15 subjects (with subject-specific normalization) would allow learning subject-invariant neural features and mitigate per-subject failure modes like the sub-26 prediction collapse.

Riemannian domain adaptation.

More aggressive covariance recentering - projecting localizer and imagery covariances to a shared reference point on the manifold - warrants further investigation.

Alpha-band revisited.

Re-evaluate with (a) time-resolved Hilbert envelopes in 200 ms windows, (b) channel-relative topographic alpha power, and (c) beta and lower-gamma bands.

Deep learning decoders.

EEGNet (Lawhern et al., 2018) and related architectures designed for EEG/MEG classification could replace feature-based classifiers, learning spatiotemporal representations directly from the raw signal.

12 Conclusion

We presented a multi-pathway ensemble approach for decoding imagined object categories from MEG brain recordings, developed as a competition entry for the IMAGINE Decoding Challenge on Kaggle (Johns Hopkins DNN 605.742, Spring 2026). The central contribution is the identification of a same-modality auditory transfer pathway: because both localizer and imagine paradigms share an

initial auditory processing phase in response to the same spoken word, training on the localizer auditory window and applying decoders to the imagine auditory window achieves near-zero-gap domain transfer. Combined with full-window multinomial classifiers and GloVe word embedding regression in a six-decoder weighted ensemble, this approach achieves leaderboard accuracy 0.133 (rank 27/169), improving 23% relative over the initial SVC baseline.

Our systematic failure-mode analysis demonstrates that the Riemannian tangent-space covariance decoder fails not due to limited capacity but due to genuine between-domain distribution mismatch, and that alpha-band log-variance features are uninformative regardless of the training domain - a diagnostic distinction with practical implications. The key lesson for neural decoding in cross-paradigm settings: domain alignment is the primary bottleneck, and identifying structural similarities between paradigms is more valuable than increasing model complexity.

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